Agenda

* Business Case : Spam vs Non Spam
* Issue with Accuracy
* Confusion Matrix
* Precision
* Recall
* F1-score

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Guess The word



?

+

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3\times 4}$$

Confusion

Matrix

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Spam or not spam

Binary classification

class 1 → Spam [positive]

class 0 → Non-spam [Negative]

⇒ Test data

class 0 | class 1

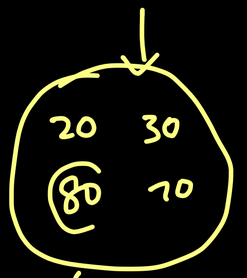
850 | 350

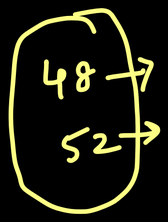
≡

Test

class 0 ≃ 50%.

1 ≃ 50%.





Not spam = $\frac{850}{1200} \times 100 = 70$

spam = $\frac{350}{1200} \times 100 \simeq 30$

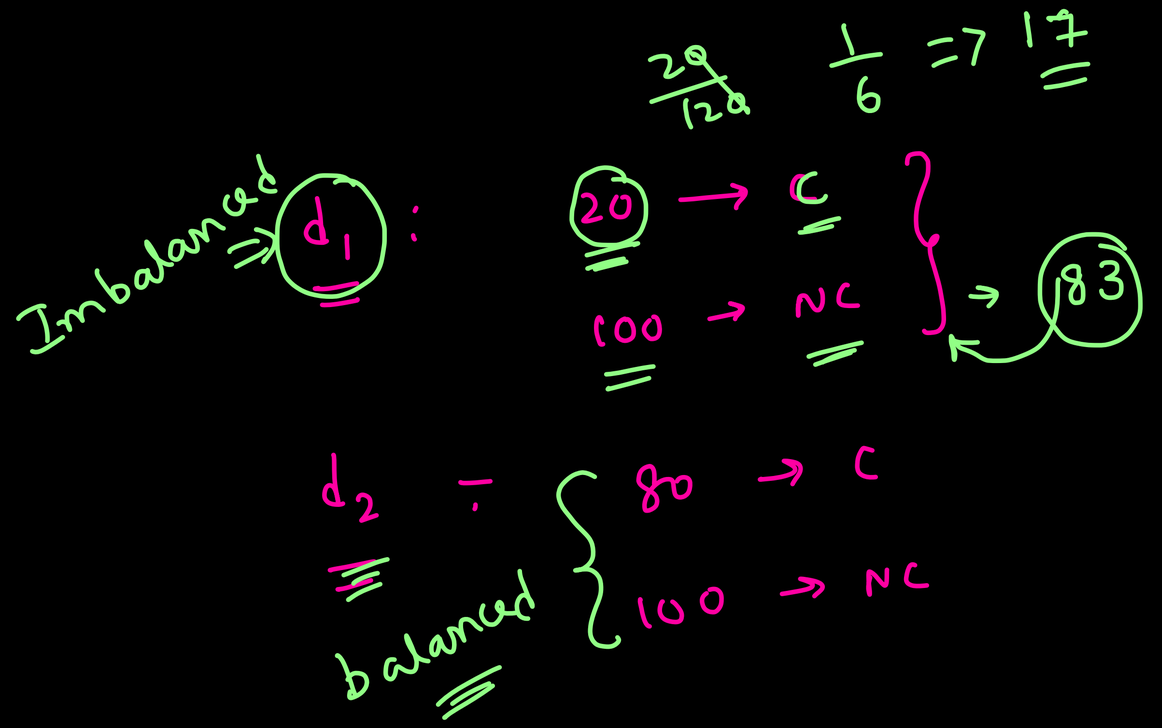
quiz-1

if data1: 20 Cancer Patients and 100 non-Cancer Patients and data2: 80 Cancer Patients 100 non-Cancer Patients, then:

1 user has participated

✓ A	Data1 = Imbalance, Data2 = balance	100%
B	Data1 = balance, Data2 = Imbalance	0%
C	Data1 = balance, Data2 = balance	0%
D	Data1 = Imbalance, Data2 = Imbalance	0%

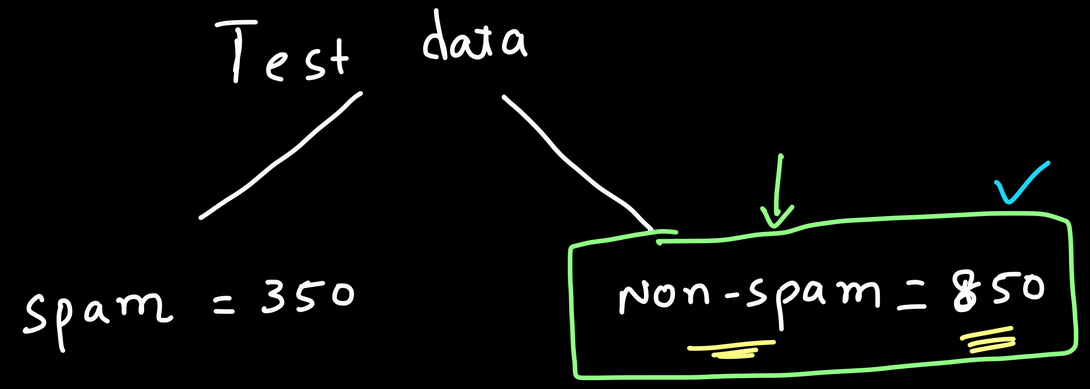
End Quiz Now



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Problem with Accuracy Metrix

Dumb Model:



Dumb model : Every Email non-spam

Accuracy = $\frac{850}{1200} \approx 70\%$

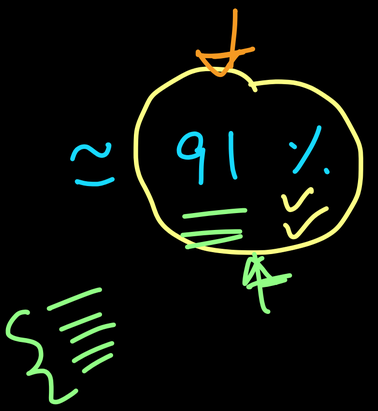
Spam → 100

Non-spam → 1100

} Imbalanced ⇒

{ Accuracy ⇒ dumb model

$\frac{1100}{1200} \simeq 91\%$

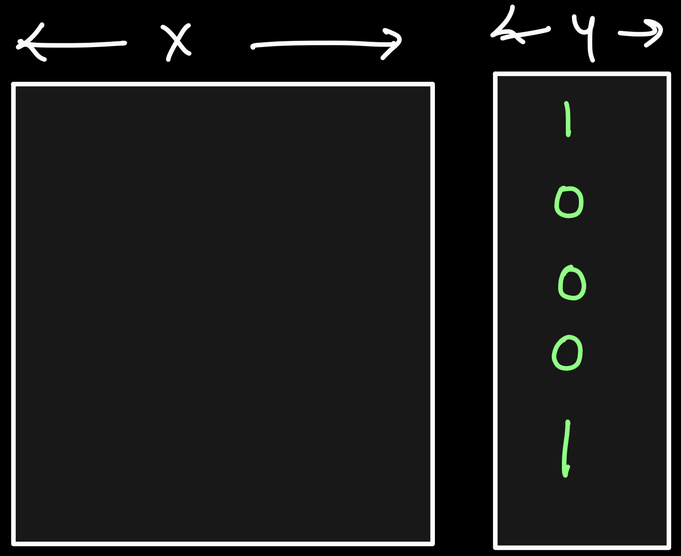


⇒ non spam ↑s , model accuracy ↑s ↓↓

Model doesn't classify spam emails. ↓↓

Issue with Accuracy

1. When data is Imbalanced ⇒ Accuracy is bad metric
2. probability is not consider



$\hat{y}_{m1}$ | label

0.92	1
0.11	0
0.07	0
0.14	0
0.95	1

$p(y=1|x_i)$

$\hat{y}_{m2}$

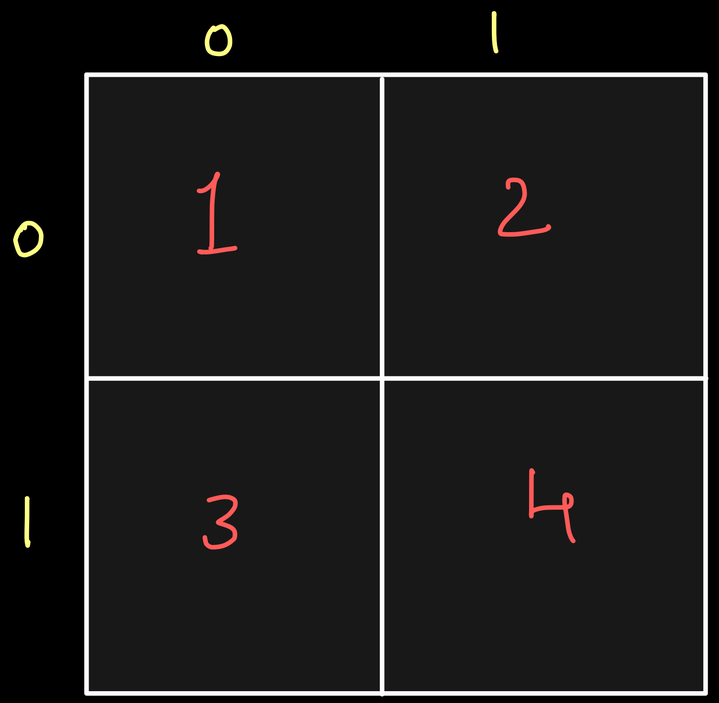
0.72	1
0.42	0
0.35	0
0.21	0
0.85	1

100% Accuracy

How to overcome the issues of accuracy.

Confusion Matrix

Measures the no of data points
1. correctly predicted in each class
2 Incorrectly " " "



1 → $y = 0$ & $\hat{y} = 0$ correct predict

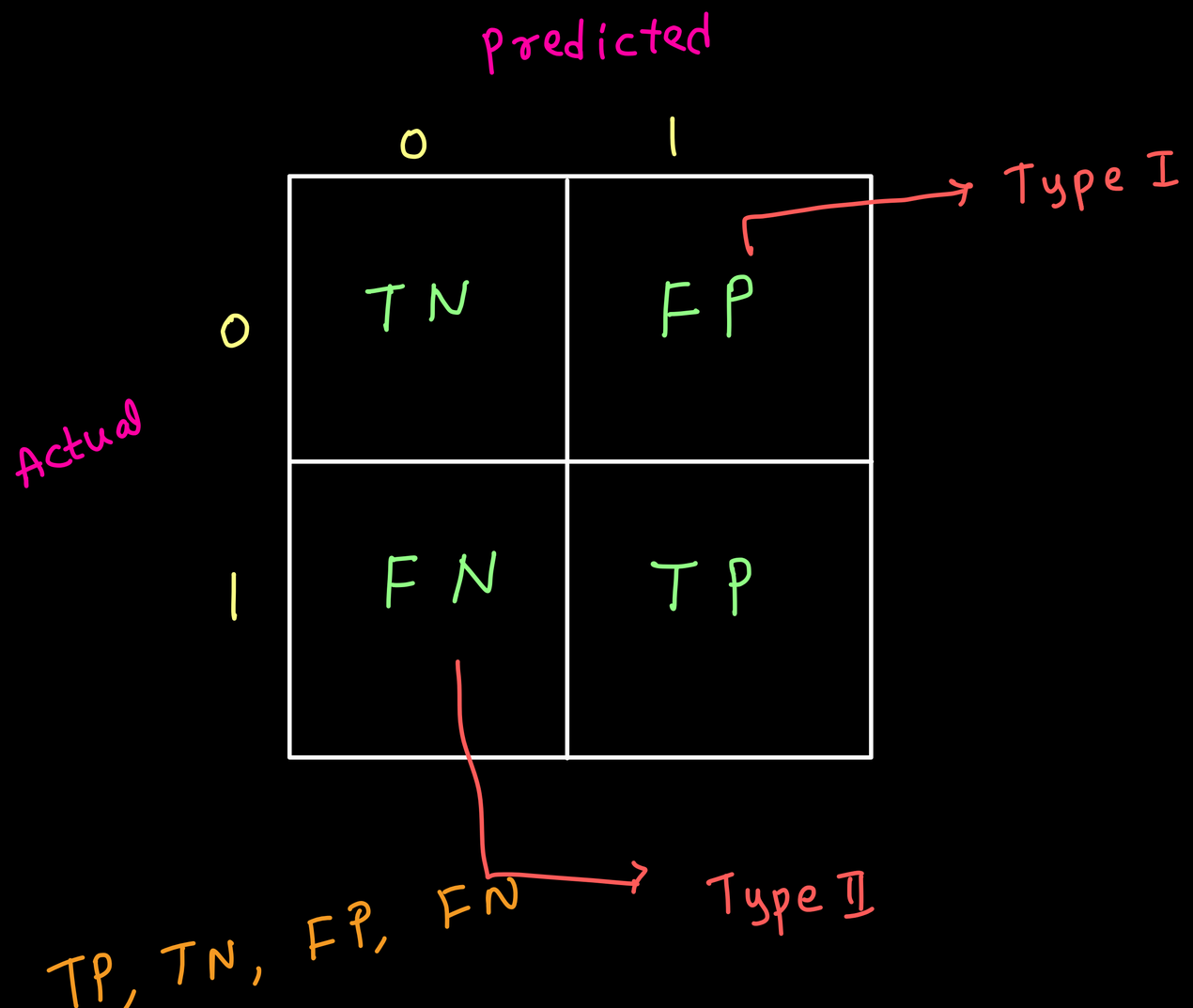
2 → $y = 0$ & $\hat{y} = 1$

3 → $y = 1$ & $\hat{y} = 0$

Incorrect prediction

4 → $y = 1$ & $\hat{y} = 1$ → correct prediction

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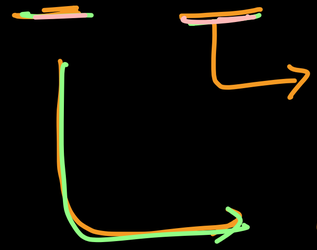
TP, TN, FP, FN

TN → $y = 0$ & $\hat{y} = 0$

TP → $y = 1$ & $\hat{y} = 1$

FN → $y = 1$ & $\hat{y} = 0$

FP → $y = 0$ & $\hat{y} = 1$



T → prediction? P/N

are you correct?

↳ T/F

Confusion Matrix for Multi class classification

	0	1	2	3
0	✓	X	X	X
1	X	✓	X	X
2	X	X	✓	X
3	X	X	X	✓

Quiz

if model classifies students into classes A,B and C , then Confusion matrix looks like ?

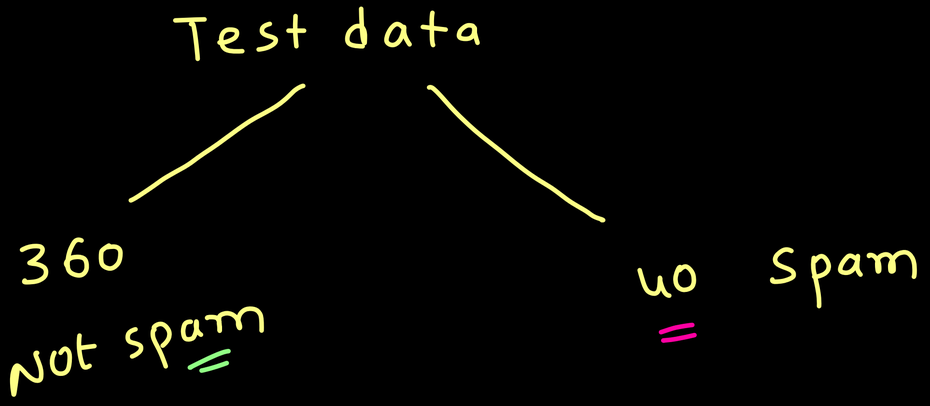
1 user has participated

A	12 x 12	0%
B	2 x 2	0%
C	4 x 4	0%
✓ D	3 X 3	100%

End Quiz Now

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understanding confusion Matrix

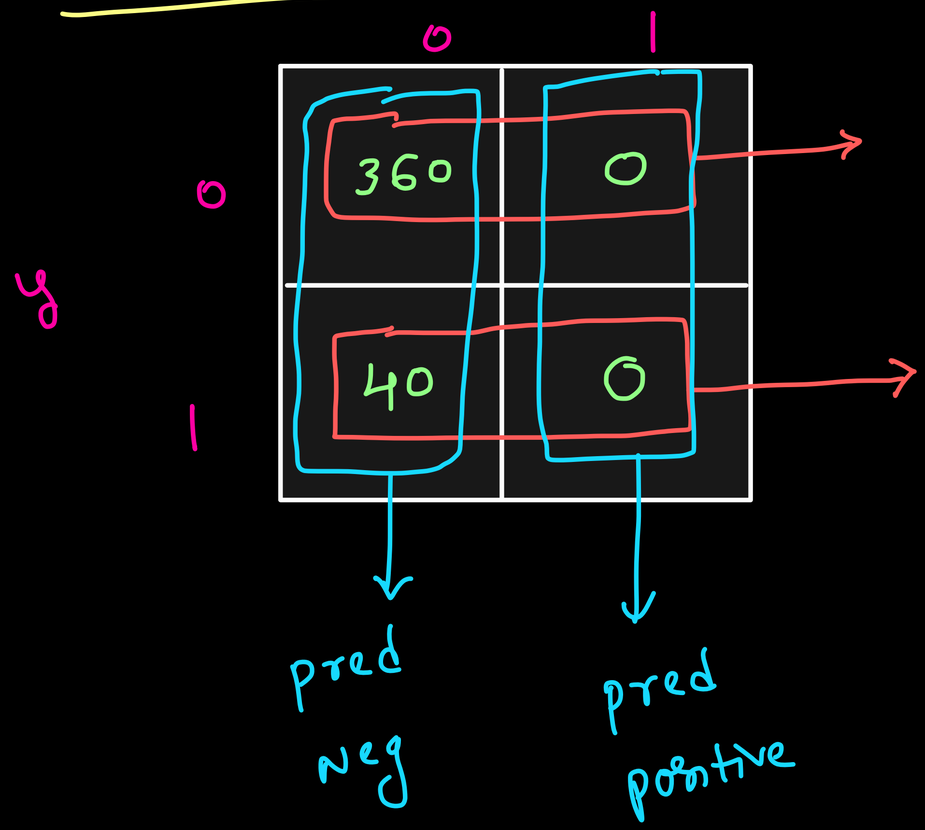


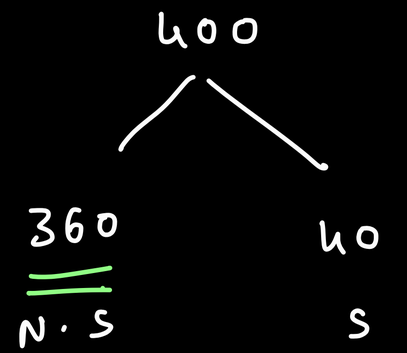
Ideal model confusion matrix

y \ ŷ	0	1
0	360	0
1	0	40

[FP, FN = 0]

Dumb model : $\hat{y}$





10 : 19 → 10 : 25

$$\text{Accuracy} = \frac{\text{Correctly Predictions}}{\text{Total samples}}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

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Quiz

For Ideal Model, which of the following is true?

1 user has participated

✓ A	FP and FN ↓, while TP and TN ↑	0%
B	TP and TN ↓, while FP and FN ↑	0%
C	TP and FN ↓, while FP and TN ↑	100%
D	FP and TN ↓, while TP and FN ↑	0%

End Quiz Now

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Precision

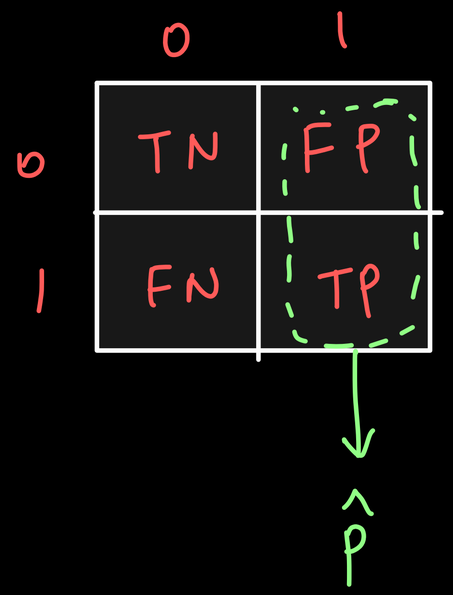
1. Receiving a spam email in inbox — FN
2. Missing out an offer letter (by categorizing it as spam) — FP

more dangerous? ✓✓

Case 2

Goal: improve TP ↑s, minimize FP ↓ } precision ≡

$$\text{precision} = \frac{TP}{TP + FP}$$



$$= \frac{TP}{\hat{p}}$$

$\Rightarrow$ out of all emails you predicted as spam, How many of emails are actually spam

$\Rightarrow$ pred 80 spam emails

Actual 74

$$\text{precision} = \frac{74}{80} = 92.5\%.$$

if FP = 0

$$\text{precision} = \frac{TP}{TP + 0} = 1 \checkmark \Rightarrow 100\%.$$

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	0	1
0	360	0
1	40	0

Test data

360
not spam

40
spam

dumb model

⇒ ~~360~~ 400

10

= 90 %

$$\text{precision} = \frac{0}{0+0+\epsilon} \simeq 0\%$$

$\epsilon \to 10^{-6}$

Ideal model

	0	1
0	360	0
1	0	40

$$\text{precision} = \frac{40}{40+0} = 1 \checkmark \Rightarrow 100\%$$

Range $\in [0, 1]$

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Recall ✓

1. A healthy patient is considered as cancerous → FP
   A cancer patient is considered as healthy → FN

Class1: cancer
Class0: N.C
2 ✓

Goal

Need a metric minimizes FN → increase TP ✓

$$\text{Recall} = \frac{TP}{TP + FN}$$

=> out of all the cancerous patients how many you are predicted cancerous

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Eg.

40 actual cancerous patients

88 predicted " "

Recall => 100 % ✓

dumb model

	0	1
0	360	0
1	40	0

$$\Rightarrow \quad \text{Recall} = \frac{TP}{TP+FN} = \frac{0}{0+40} = 0 \checkmark\checkmark$$

Ideal model

360	0
0	40

$$\text{Recall} = \frac{40}{40+0} = 1 \checkmark\checkmark$$

$$\text{Recall} \in [0, 1] \checkmark\checkmark$$

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→ if your job is to reduce FN → Recall

" " " FP → precision

F1-Score

Task. Credit card transcation (fraud) (or) legitimate

1. predicting a transcation as legit when it is actually fraud
   ↳ leads to finantial loss FN

2 predicting transcations as fraud when it is legit
(inconvenience to card holder) FP

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⇒ Goal: FP & FN ↓ → TP ↑

	Precision	Recall	Arithmetic mean avg	F1-Score
M_1	0.3	0.8	0.55	$\frac{2 * 0.3 * 0.8}{0.3 + 0.8} = 0.43$
M_2	0.9	0.2	0.55	0.32
M_3	0.7	0.4	0.55	0.5
M_4	0.1	1	0.55	0.18

M_3

a, b

$$A.M = \frac{a+b}{2}$$

$$H.M = \frac{2}{\frac{1}{a} + \frac{1}{b}}$$

Harmonic mean

F1-score = H.M of (P, R) (P: prec, R: Recall)

$$= \frac{2}{\frac{P+R}{PR}}$$

$$\text{F1-score} = \frac{2PR}{P+R}$$

Why does the F-1 score use Harmonic Mean (HM) instead of Arithmetic Mean (AM) ?

1 user has participated

	A	AM penalizes models the most when even Precision and Recall are low.	0%
✓	B	HM penalizes models the most when even Precision and Recall are low.	0%
	C	HM penalizes models the most when even Precision and Recall are high.	100%
	D	AM penalizes models the most when even Precision and Recall are high.	0%

End Quiz Now

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dumb model

$$F1 = \frac{2 \cdot P \, R}{P + R} \geq \frac{2 \cdot 0 \cdot 0}{0 + 0} = 0$$

Ideal model

$$F1 = \frac{2 \cdot 1 \cdot 1}{1 + 1} = 1 \checkmark$$

$$F1 \in [0, 1]$$

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